Design and Validation of a Modular Quick-Connect Prosthetic Foot Interface

Interim Technical Report

1. Objective

Current prosthetic pyramid joints are used as a mechanical connection between the limb and the foot and are aligned using a single four screwed connection by a technician. As both functions of the interface share the same connection, a patient cannot trial or swap feet at home without risking disturbance of the alignment. The objective of this project is to design a modular interface that separates these two functions, allowing a technician to set alignment once per trial foot while giving the patient a repeatable, tool-free quick-connect for routine swaps. The interface is being sized to a 70 kg male subject, with boundary conditions derived from an earlier anthropometric scaling exercise based on a 95th-percentile USA male of 188 cm and approximately 112 kg.

2. Determination of Boundary Conditions

Ground reaction forces, moments about the ankle joint, segment inertial properties and centre of mass locations were required at discrete points through the gait cycle. These were obtained by sampling Winter's dataset and scaling it to the 95th-percentile subject using a force, length and moment scaling factor:

SFF = (95th-percentile USA male mass) / (Winter's reference mass)

SFL = (95th-percentile USA male height) / (Winter's reference height)

SFM = SFL × SFF

Where SFF is the Force Scaling Factor, SFL is the Length Scaling Factor and SFM is the Moment Scaling Factor.

Applying these factors to Winter's tabulated data produced a scaled boundary condition dataset, including a ground reaction force profile across the stance phase, from which discrete loading cases were later extracted for finite element analysis. For the 70 kg subject-specific stage of the project, it remains to be confirmed whether the boundary condition values already stored in the working dataset are normalized per kilogram of body mass or already converted to absolute Newton and Newton-metre values, since this determines how the substitution from the 112 kg reference subject to the 70 kg target subject should be carried out. This is treated as an open item rather than assumed, consistent with the general requirement that every boundary condition value be traceable to a source rather than estimated.

3. Characterisation of the Existing Pyramid Joint

Before redesigning the interface, the deflection angle tolerance of the current pyramid joint was characterised, since this value sets the baseline that any replacement joint must match or improve on. The convex and concave pyramid connectors were reverse engineered in SolidWorks to ISO 24562, using the sketch, revolve and extrude functions; a circular pattern was used to generate the slanted screw holes on the concave connector. The angular tolerance was then assessed in assembly by mating the pyramid centre point of the convex connector to the centroidal axis of the concave connector and applying collision detection to stop relative motion at maximum tilt. The measure tool was used to extract the resulting angular displacement between the two centroidal axes, providing a maximum tilt of approximately 6°.

This computational estimate was cross-checked against a physical experiment in which the concave connector was pushed to its extreme positions on the convex connector while mounted on a prosthetic foot. Results across two feet, with and without the retaining screws, are summarised in Table 1.

Table 1. Measured angular deflection of the existing pyramid joint.

Configuration	Front	Back	Right	Left
Foot 1, without screws	0°	-12°	-7°	-8°

Foot 1, with screws	-1°	-12°	-8°	-8°
Foot 2, with screws	0°	-13°	-6°	-7°
Foot 2, without screws	-1°	-13°	-7°	-9°

The measured deflections show a consistent asymmetry between the anterior-posterior direction (0° to -13°) and the medial-lateral direction (-6° to -9°), and only a small change between the screwed and unscrewed conditions. This indicates that the achievable tilt in the existing joint is governed primarily by geometric clearance between the pyramid and its receiver rather than by fastener preload, and that any replacement quick-connect stage should not be assumed to inherit a single isotropic tolerance value.

4. Modular Quick-Connect Interface Design

The tolerance study above supported the core design decision of this project: separating the single existing pyramid joint into distinct mechanisms, each responsible for a single constraint, rather than one joint carrying alignment, rotation and retention together. The interface has since been developed as a modular stack with three degrees of freedom.

Dowel Pin Slots

A key part of the following design is the concept of using dowel pin slots as a locating mechanism for the plates in the interface. In having two small, circular extrusions on the faces of the surface of the foot, a two-pin index is implemented, locating the discrete horizontal plate above without requiring any screws. The holes receiving the pins consist are of two different types One round hole, precisely sized to fit the dowel pin diameter, is used as the prime locator for the next layer of the interface. It removes two DOFs (X and Y in the plate's plane) and fixes the exact rotational index position. The second hole is a slot, orientated pointing towards the centre of the piece. This removes the remaining DOF without interfering with the first holes positioning. If both features were round holes, any stack-up in bolt-circle radius, pin spacing, or thermal expansion would force the pins to fight each other, resulting in an over-constrained mechanism. Both the top surface of the foot, and the discrete horizontal plate have these slots, enabling the location of the discrete horizontal and vertical plates without over constraining the assembly.

Dowel pin location

The dowel pin location on the surface of the discrete horizontal plate defines the angle of horizontal rotation the user wishes to have of the foot. It is placed at 0º SW of the centre of the interface for 0º of horizontal rotation. The part is interchangeable, with a different position of the pin corresponding to a different degree of horizontal rotation. This allows for parts to be made in the range of ±20º. The intervals of the dowel pin position can be decided by the consumer to meet their requirements.

Discrete vertical piece

This piece has three wide slots, evenly spaced along its circular base. This enable is to be independent of the discrete horizontal piece. This is as the slots cover the potential ±20º of the pin, allowing the screws to still pass through the slots and holes, locking the pieces to the surface of the foot.

Rod connecting Piece

The rod connecting piece has an inner diameter the matches that of the cylinder extrusion from the slanted, discrete vertical rotation piece, allowing the two to be tightly connected. A hole in both cylinders, through the parts, is used to receive a rod, locating the rod connecting piece and ensuring correct alignment. The rod connecting piece is a concept for a potential connection between the patients limb socket and the foot interface. This rod can vary in size as long as the end has the simple hole that can align with that of the interface's extrusion.

Thus, the interface is comprised of four main parts; the top surface of the prosthetic foot, the discrete horizontal and vertical rotation plates and the rod connecting piece. Earlier designs are documented in the appendix, depicting the design process and convergence on the final design

6. The Prosthetic Foot

The dimensions for the prosthetic foot for a 70kg male, assuming average height of 1.7m, had to fit in the dimensions governed by the following equations:

Foot Length = 0.152H = 93.5mm

Foot Breadth = 0.055H= 258.4mm

Foot Height = 0.039 H = 66.3mm

The dimensions of the foot were then scaled from Shay and Tomer's design and modified to fit perfectly within the dimensions required. The top surface of the foot was further modified to mate correctly with the base, discrete horizontal rotation plate, thus ensuring the interface fit smoothly on top of the foot.

7. Finite Element Analysis

An FEA was conducted on the prosthetic foot part in SolidWorks. The material selected for the part was the plastic Delrin 2700 from the SolidWorks directory as its Elastic Modulus of 2900MPa was the closest fit to the PLAs 3056MPa. The fixed geometry of the part was the edge of the holes for the screws and the extra surrounding surface calculated using Shigley's Frustrum approach. Two separate static FEAs were taken, one where the force was applied in plantarflexion, and one in dorsiflexion. The force was calculated to be 824N using Winter's formula of GF = 1.2BW. this was located at 38.7mm from the heel (0.15L) for plantarflexion and 190.92mm (0.74L) in dorsiflexion. The mesh size was set to a standard mesh with a Global Size 3mm, and a Tolerance of 3mm.

Desired results:

Angle of deflection in plantarflexion: 5º

Angle of deflection in dorsiflexion: 10-15º

Stress margin: <42MPa

The dimensions of the foot were adjusted to satisfy the requirements in plantarflexion but failed to do sin dorsiflexion.

8. Discussion

Taken together, the tolerance characterisation and the FEA methodology give two independent routes to validating the redesigned interface: a geometric, assembly-based estimate of achievable deflection, and a load-based finite element estimate of deformation and strain energy under gait loading. Agreement between the two would support the current interface architecture; disagreement would indicate that either the assumed contact geometry or the applied boundary conditions require revision. The unresolved question of whether the stored boundary condition data is normalized or absolute is treated as a blocking item for the 70 kg loading case, since applying the wrong assumption would silently scale every downstream result by the ratio of the two reference masses.

8. Conclusion and Future Work

The project has progressed from characterising the tolerance of the existing single-joint pyramid connector to a modular three-degree-of-freedom interface that separates alignment, rotation and retention into distinct features, together with an FEA methodology adapted from Tomer and Shay's report to the present 70 kg adult case. Remaining work includes finalising the sketch dimensions for the prosthetic foot to satisfy the conditions required in dorsiflexion, to create and print the prosthetic foot and to trial the interface, foot and rod connection on a real patient 70kg to obtain real world data.

9. Appendix

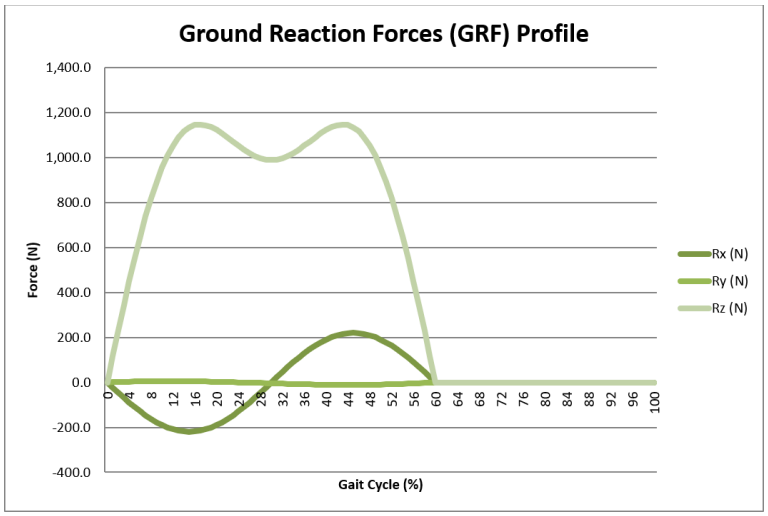


Figure (1): A plot of the Ground Reaction Forces extrapolated for the 95% height and weight USA male using Winter's dataset

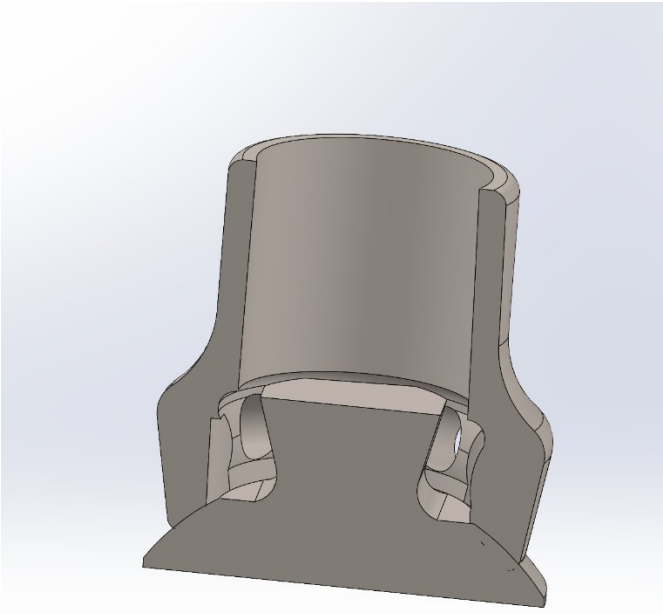


Figure (2): The original ISO 24562 assembly of the convex and concave parts modelled in SolidWorks

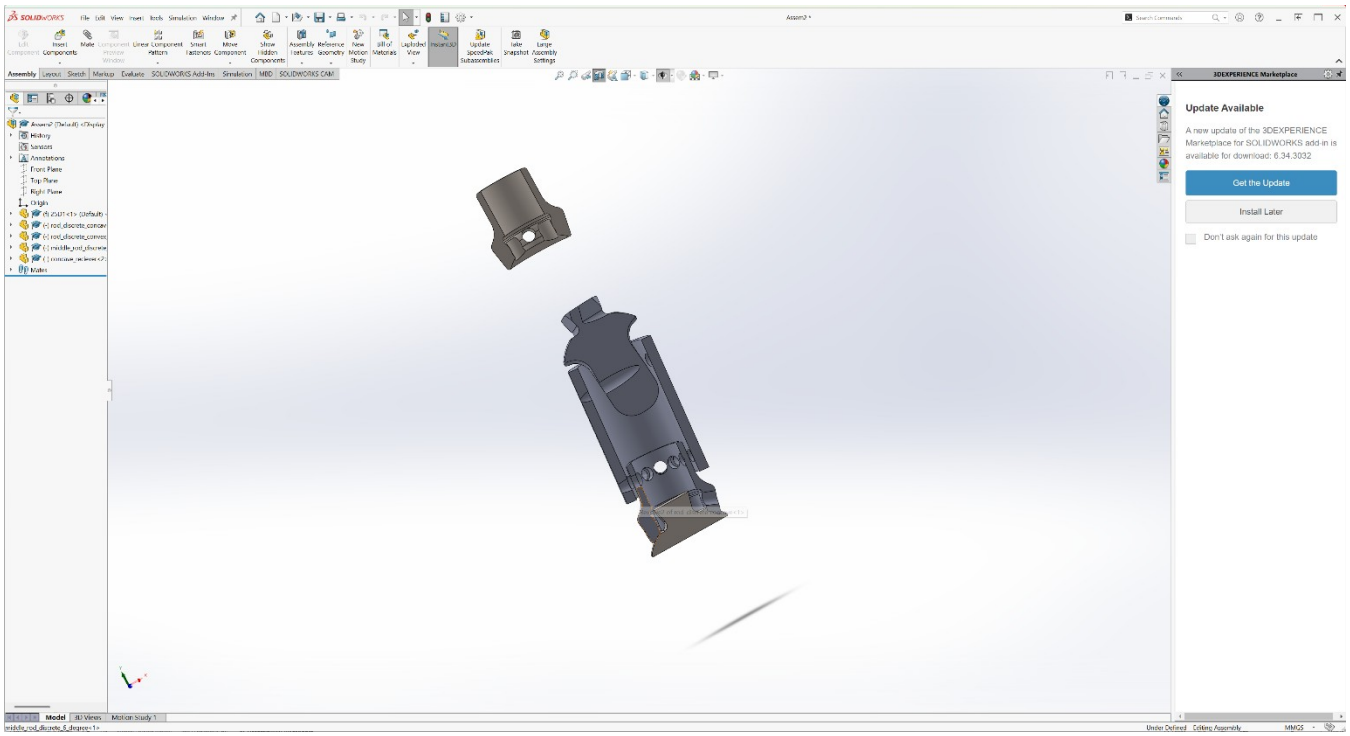


Figure (2): The first attempt at obtaining discrete rotation in 3 DOFs, utilising the pyramid joint

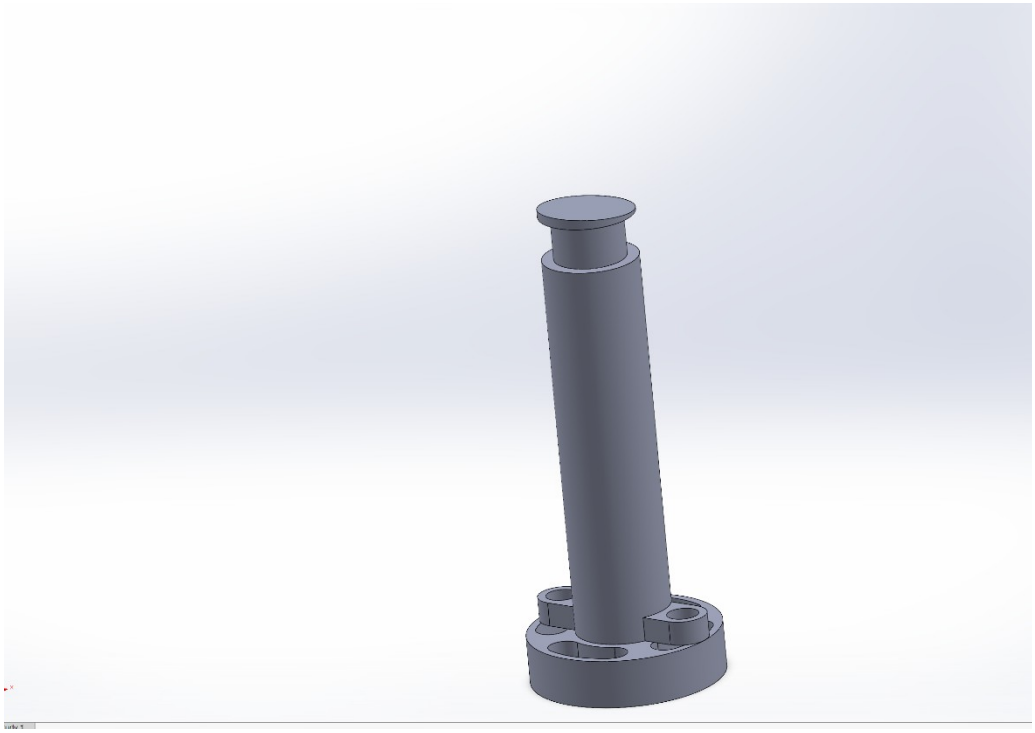


Figure (3): First usage of the wide slots to enable continuous horizontal rotation, and the usage of an adjustable wedge for a discrete vertical angle.

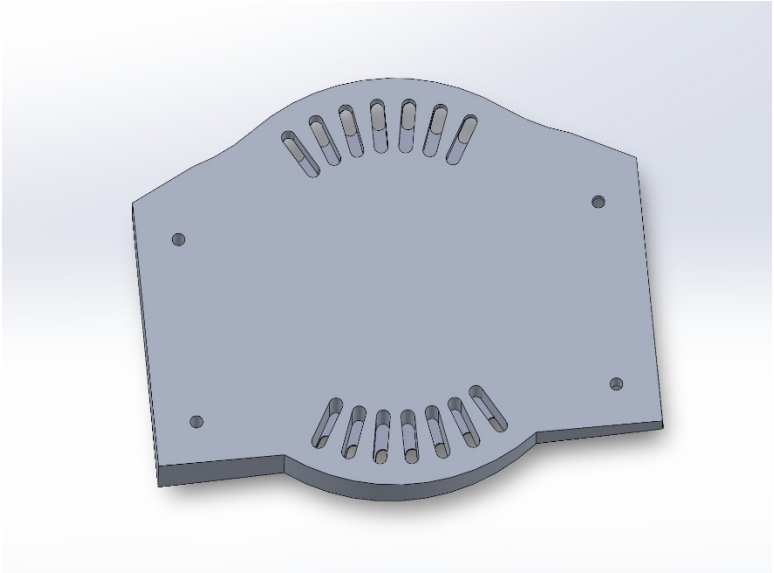


Figure (4): First attempt at matching the interface to the desired surface pf the foot, utilising slots.

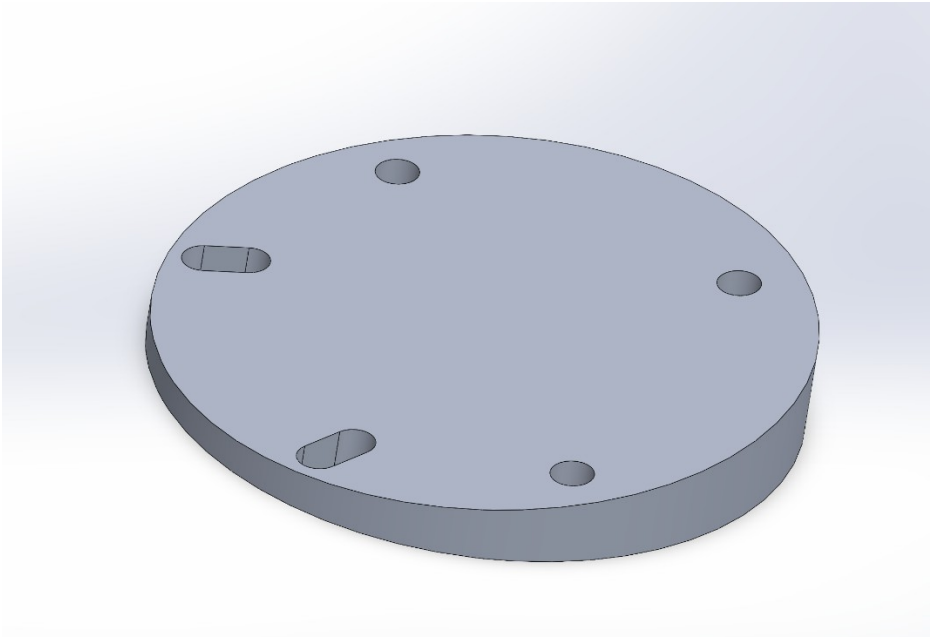


Figure (5): Converging on the minimalist, slot approach

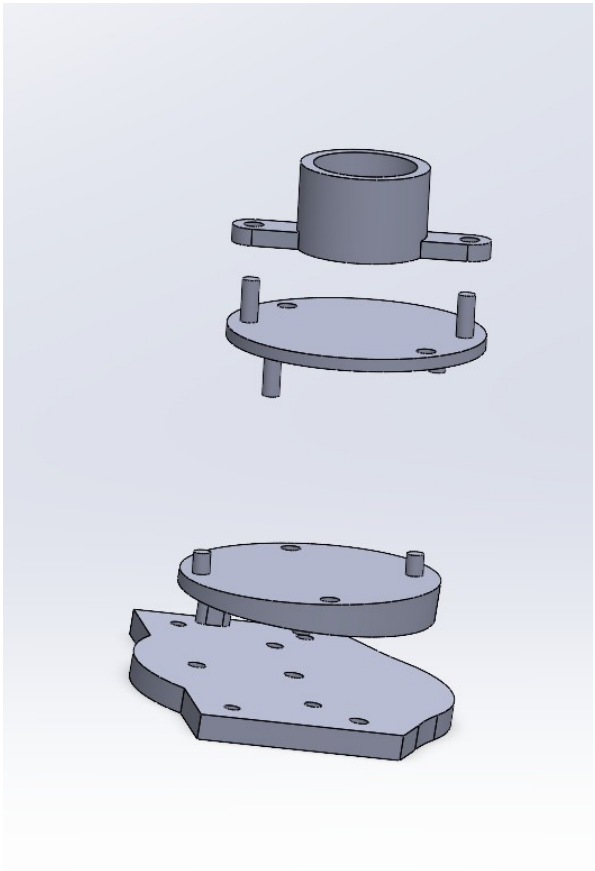


Figure (6): The dual pin attempt, which failed due to its inability to be screwed down.

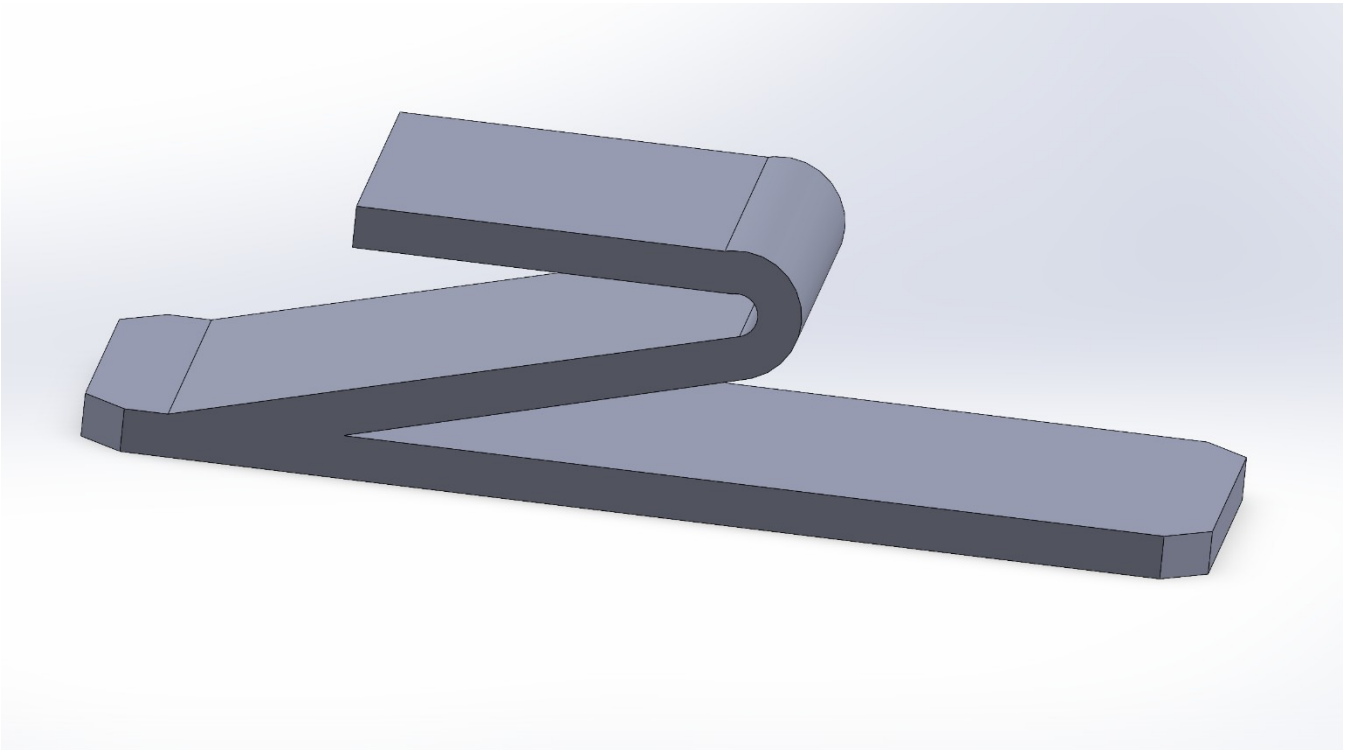


Figure (7): The first model of the prosthetic foot, fitting Winter's dimensions

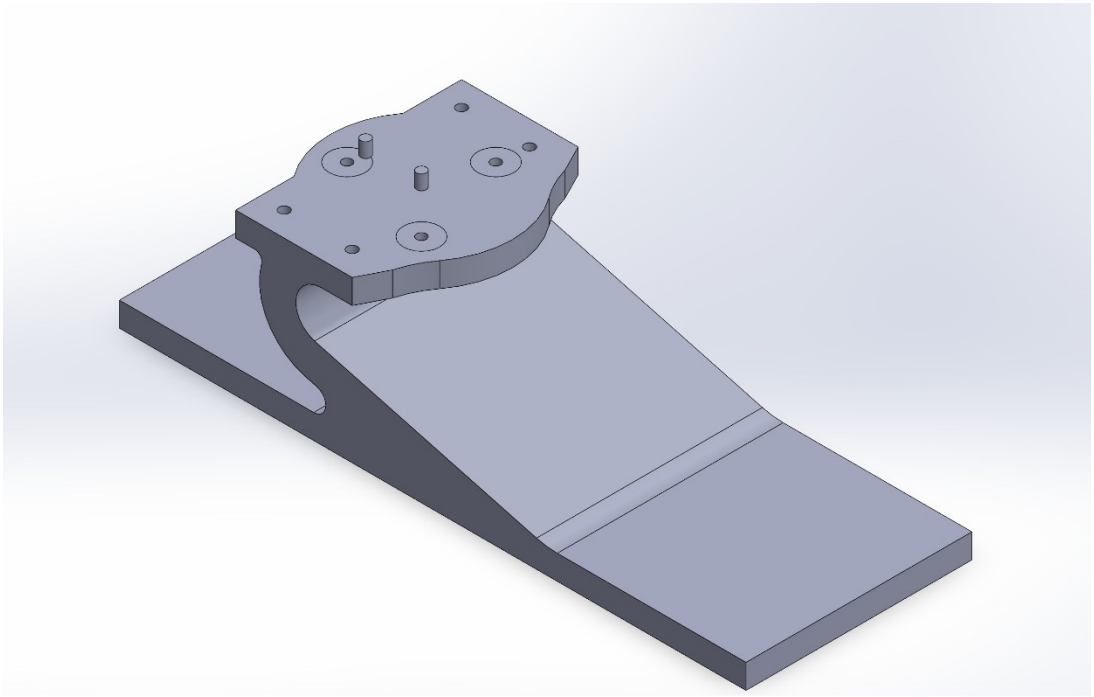


Figure (8): The foot design that underwent FEA testing, satisfying plantarflexion conditions but not dorsiflexion

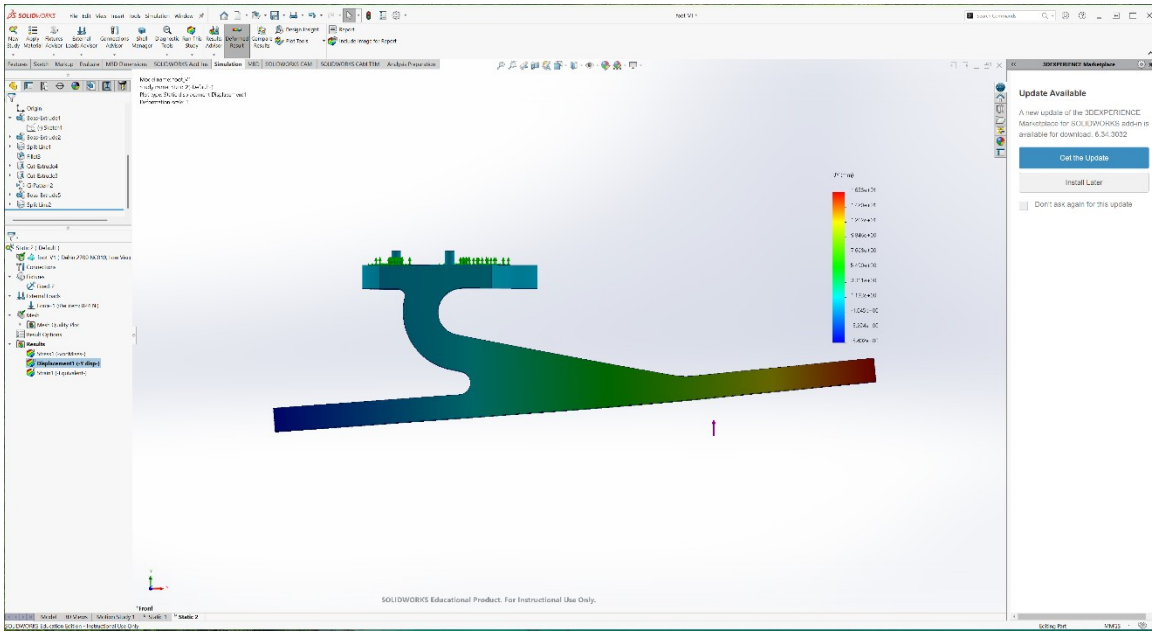


Figure (9): The displacement test results of the prosthetic foot in dorsiflexion

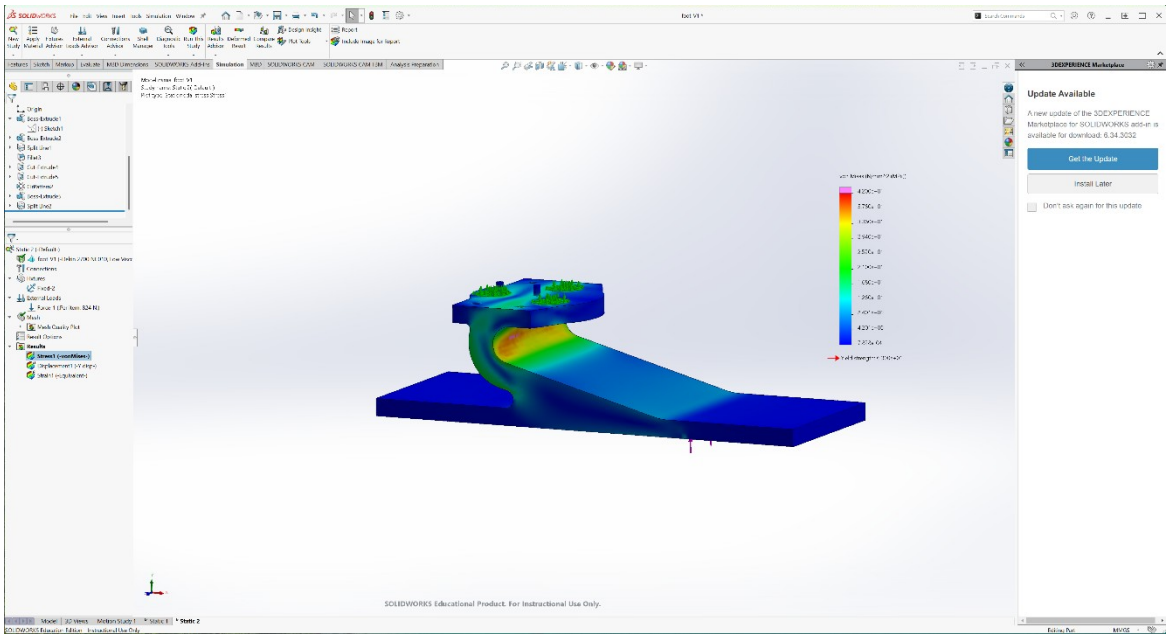


Figure (10): The stress map of the prosthetic foot in dorsiflexion

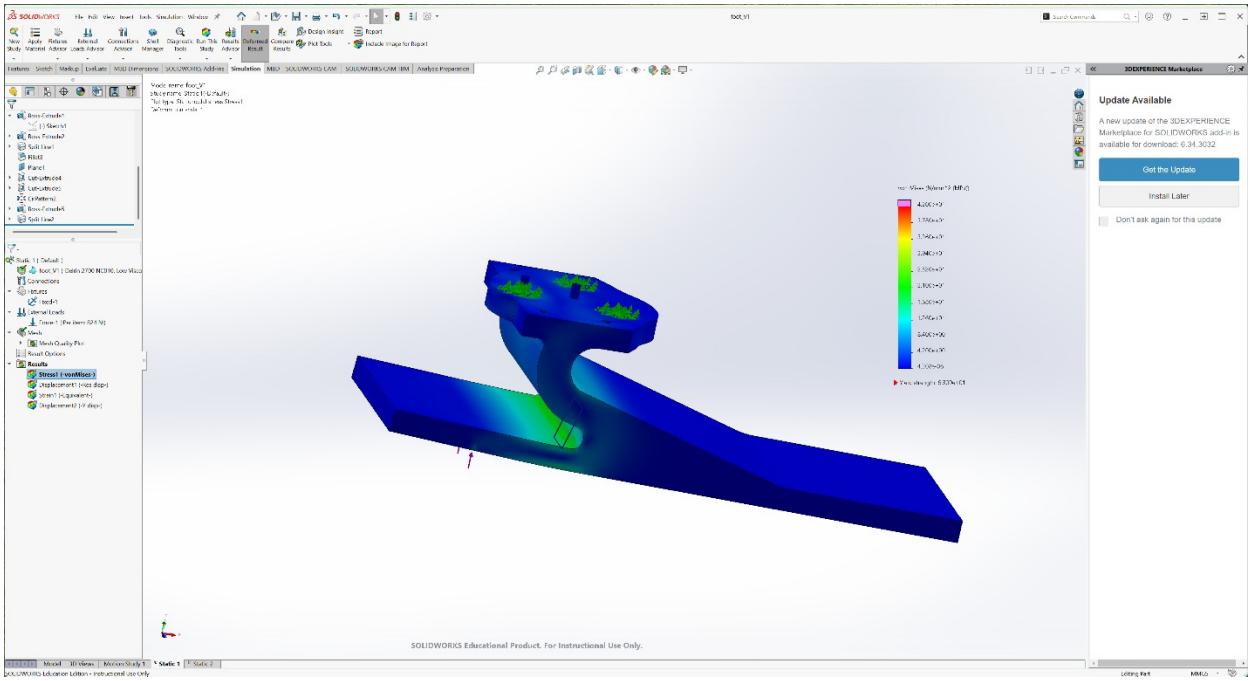


Figure (11); The stress map of the prosthetic foot in plantarflexion

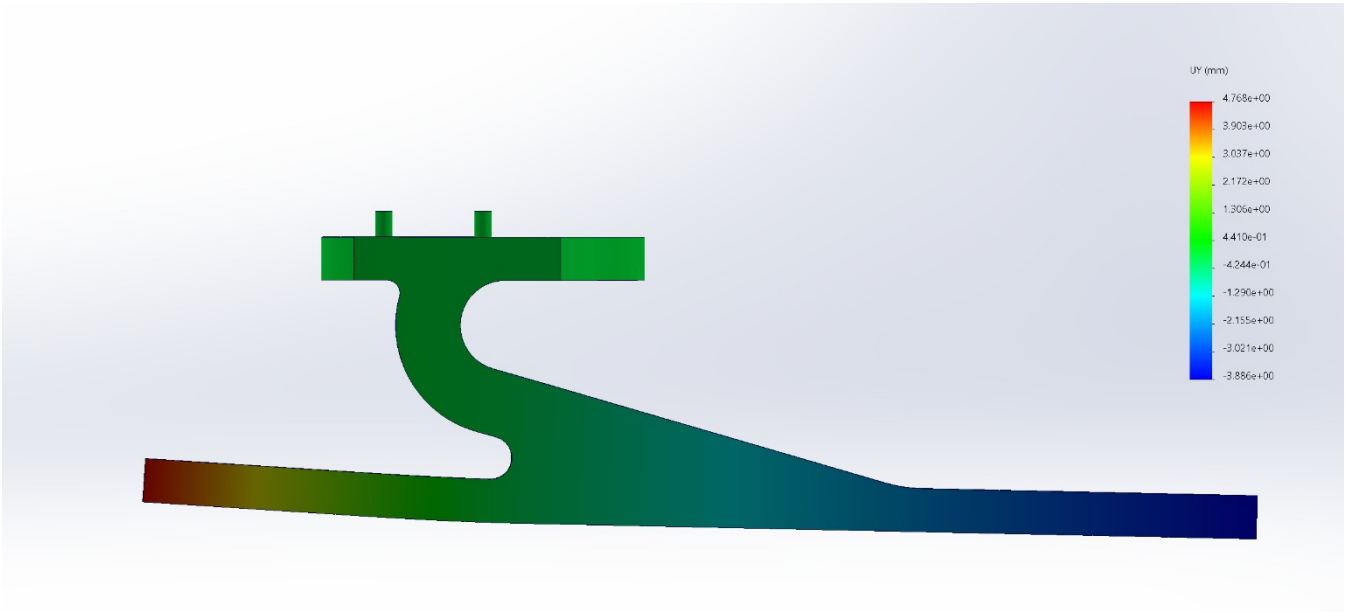


Figure (12): The stress map of the prosthetic foot in plantarflexion

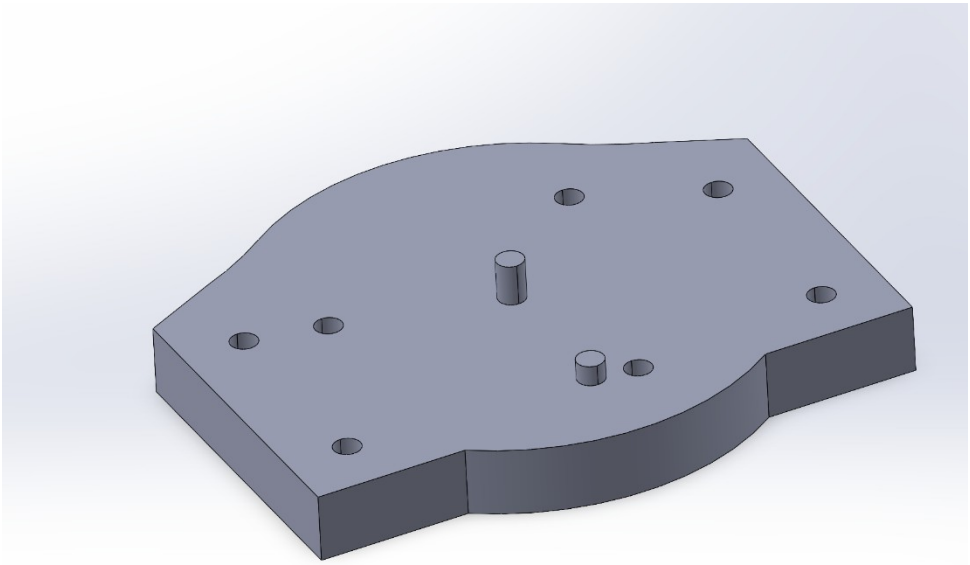


Figure (13) : The top plate of the prosthetic foot

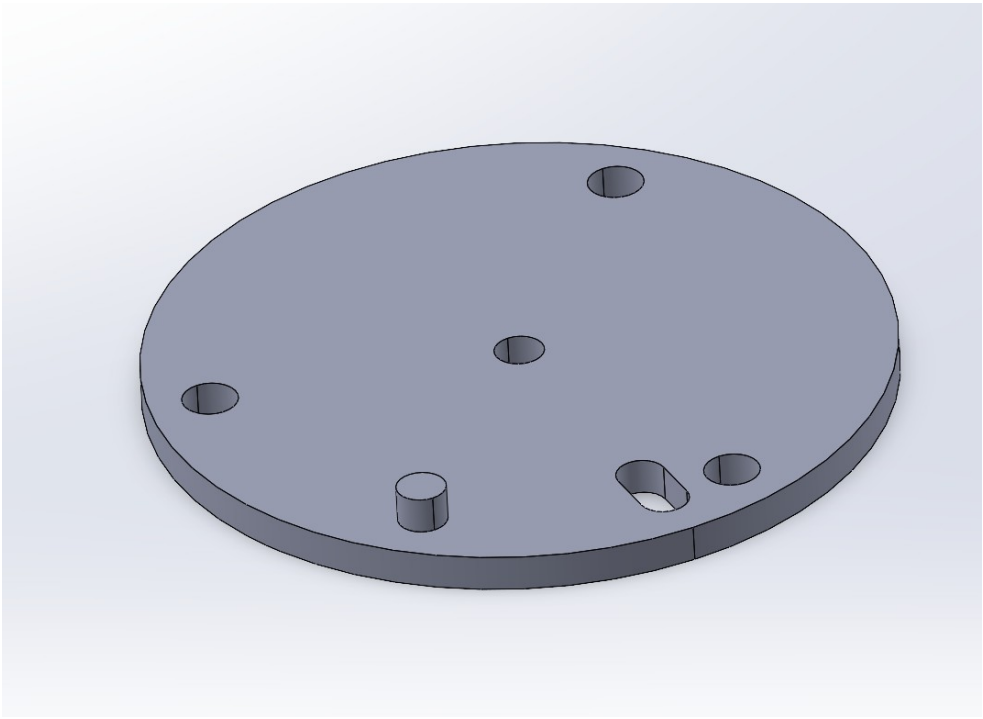


Figure (14): The discrete horizontal rotation plate

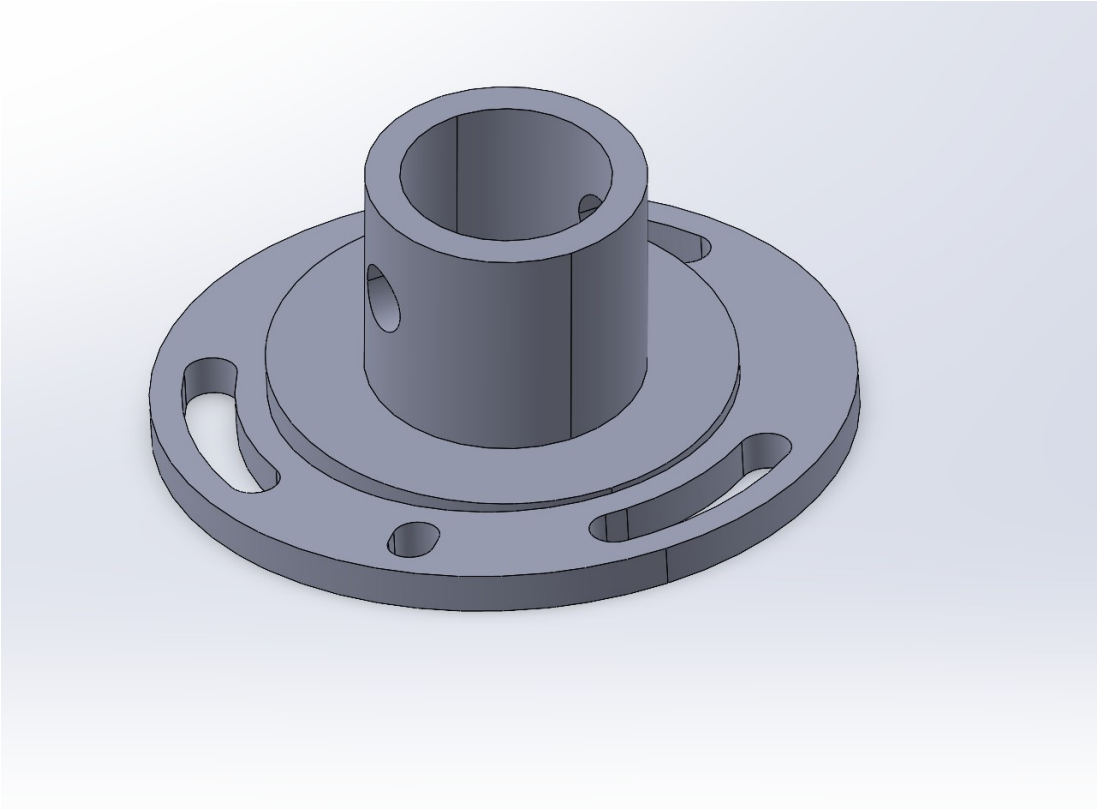


Figure (15): The discrete vertical rotation plate

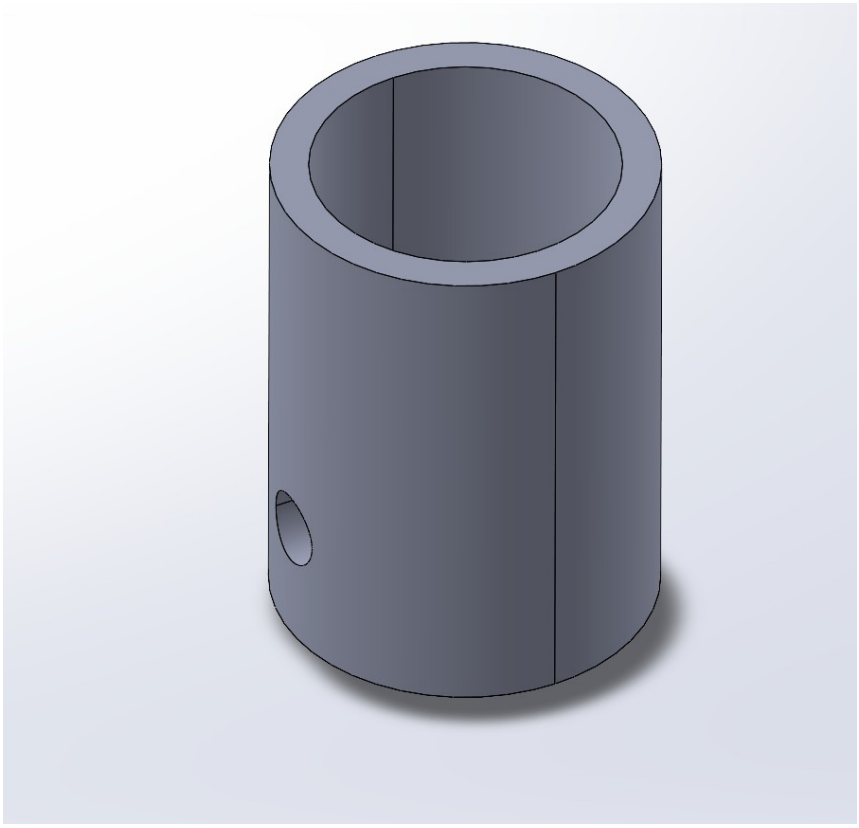


Figure (16): The rod connecting piece

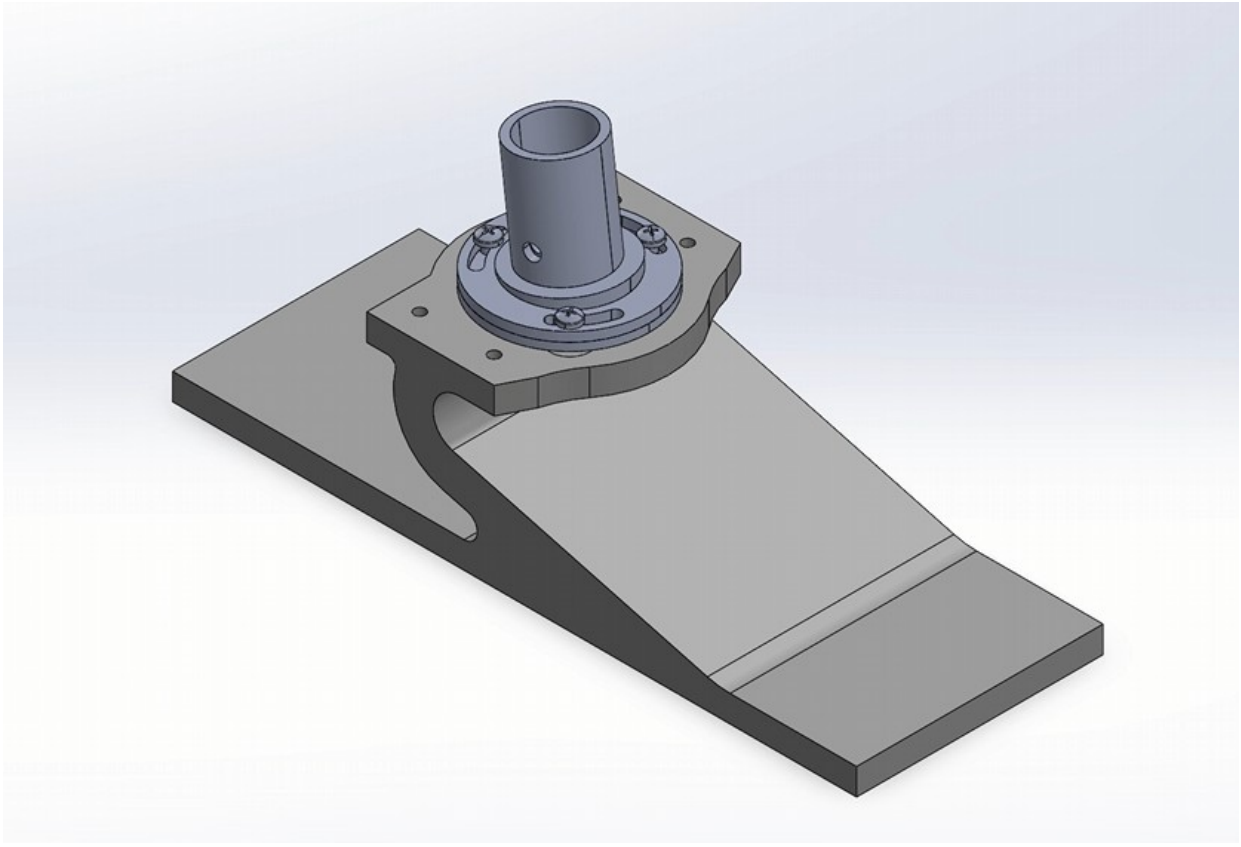


Figure (17): The interface and foot assembly